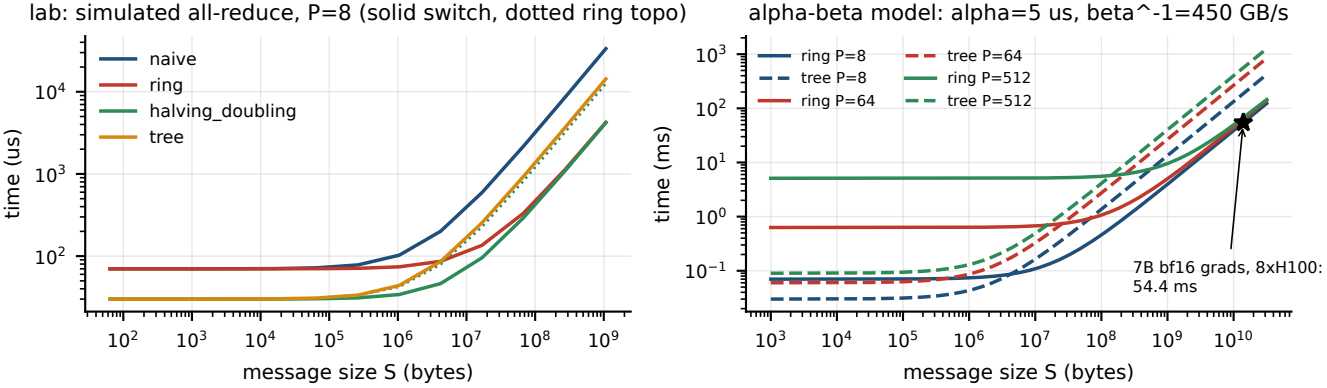
lab: simulated all-reduce, P=8 (solid switch, dotted ring topo)
naive
ring
halving_doubling
tree
time (us)
message size S (bytes)
alpha-beta model: alpha=5 us, beta^-1=450 GB/s
ring P=8
tree P=8
ring P=64
tree P=64
ring P=512
tree P=512
time (ms)
7B bf16 grads, 8xH100:
54.4 ms
message size S (bytes)